CONNECTED DEVICES • FINAL PROJECT

Pet Activity & Proximity Tracker

A wearable sensing tag for pets and a tabletop display
that shows daily activity and companionship — calmly, without alerts.

🔗 **GitHub:** https://github.com/MengqiS21/TECHIN514_FinalProject

ESP32-C3 | BLE | LSM6DS3 IMU | Stepper Motor

The Problem

😔 You don't know what your pet does all day

Activity patterns, rest time, and energy levels are invisible when you're away or busy.

📍 No easy way to measure togetherness

There's no simple tool to know how much time your pet actually spends near you each day.

🔔 Existing trackers are too noisy

Most pet trackers focus on GPS and push notifications, they're designed for anxiety, not reflection.

🔋 Wearables must be small & light

A pet collar tag needs to be compact, battery-friendly, and simple, no GPS, no cloud.

Proposed Solution

Collar Sensing Tag

Worn on the pet's collar. Detects movement and estimates proximity to the display.

- ESP32-C3 — main controller + BLE
- LSM6DS3 IMU — 3-axis accelerometer
- MCP1700 LDO — 3.3V from LiPo
- 3.7V LiPo battery
- Classifies activity into a 0–100 score
- Sends score to display via BLE

Bond Meter Display

Sits on a table at home. Shows the pet's daily activity score using a gauge needle.

- ESP32-C3 — main controller + BLE
- 28BYJ-48 stepper motor — gauge needle
- WS2812 RGB LED — proximity feedback
- 3.7V LiPo or USB 5V power
- Receives BLE data from collar tag
- Button to reset or change mode

System Architecture

Collar Tag (ESP32-C3)

- LSM6DS3 IMU
- MCP1700 LDO
- 3.7V LiPo
- BLE Radio

BLE

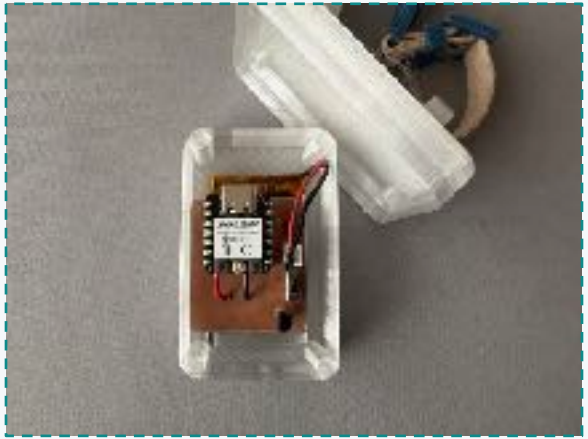
Activity Score
(0–100)

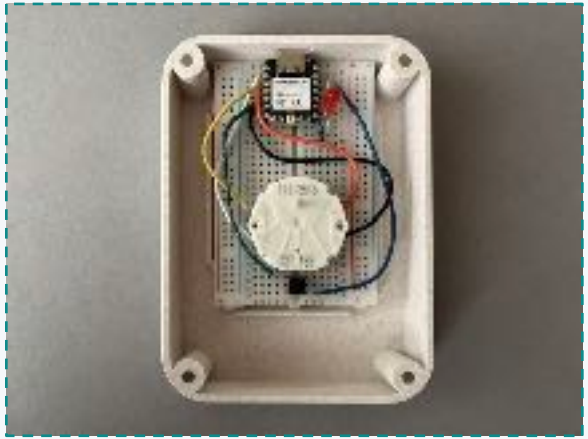
Bond Meter (ESP32-C3)

- Stepper Motor(x27 168)
- LED
- USB 5V
- Button

No cloud • No GPS • No phone required

Critical Components & PCB





Component	Part	Device	Notes
MCU + BLE	ESP32-C3	Both	Main controller
IMU (Accel)	LSM6DS3	Collar Tag	I2C, 3-axis
LDO Regulator	MCP1700	Collar Tag	3.3V, low quiescent current
Stepper Motor	28BYJ-48	Bond Meter	Gauge needle driver
RGB LED	WS2812	Bond Meter	Proximity feedback
Battery	3.7V LiPo	Both	Collar: ~150–300mAh / Display: ~500–1200mAh

Signal Processing

How it works (on the collar tag):

1. **Sample accelerometer**
   LSM6DS3 sampled every 40 ms over a 1.5 s window
2. **Compute magnitude**
   |ax| + |ay| + |az| for each sample
3. **Average the window**
   Mean magnitude over the 1.5 s window
4. **Map to 0–100 score**
   Linear mapping: 0g → 0, 3g → 100
5. **Send via BLE**
   Score sent to display for real-time gauge update

Accuracy

No formal evaluation yet.
No labeled dataset or test metrics (accuracy, precision, recall) have been collected.

Formal accuracy: N/A

Design Notes

No ML model needed
Simple threshold mapping works well for this use case.

No GPS or cloud
BLE RSSI is used for proximity — all processing is on-device.

Battery Considerations

🐾 Collar Tag

3.7V LiPo • ~150–300 mAh

- Deep sleep between BLE transmissions
- Wakes to sample every ~1.5 s
- BLE advertise-only (no scanning)
- MCP1700 LDO — very low quiescent current
- Target: several days per charge

📊 Bond Meter

3.7V LiPo or USB 5V • ~500–1200 mAh

- Stepper motor powered off between updates
- BLE scanning only when needed
- LED dims when pet is away
- Can use USB power for always-on use
- Target: 3–7 days on battery

⚡ General Strategy

Shared design approach

- Aggressive duty cycling on both MCUs
- No continuous data streaming
- Batch updates only (once per ~1.5 s)
- No GPS, no cloud, no WiFi

Budget Summary

Component	Part	Qty	Unit Cost	Total
MCU (ESP32-C3)	Seeed Studio ESP32-C3 XIAO dev module	2	~$10.00	~$20.00
IMU	LSM6DS3 module	1	~$7.00	~$7.00
LDO Regulator	MCP1700	1	~$0.95	~$0.95
Stepper Motor	X27 168	1	~$8.00	~$8.00
RGB LED	WS2812	1	~$0.10	~$0.10
I2C Pull-up Resistors	4.7kΩ or 10kΩ ×2	1 set	~$0.10	~$0.10
220Ω Resistor	Standard	1	<$0.10	<$0.10
Batteries	LiPo ×2 + connectors	—	~$8.99	~$17.98
PCB Fabrication	JLCPCB / PCBWay (×2 boards)	—	~$10.00	~$10.00
Misc (switch, connectors, enclosure)	—	—	—	~$8.00
TOTAL ESTIMATED				**~$72.23**

** All costs estimated. Final BOM depends on supplier and order quantity.*

Future Work

🤖 Collect labeled data & evaluate accuracy

Build a labeled dataset and measure precision/recall for activity classification.

📱 Mobile companion app

Simple BLE app to log daily summaries and show activity trends over time.

☀️ Longer battery life

Tune sleep cycles and explore solar or wireless charging for the display device.

🔒 Better enclosure

Weatherproof, pet-safe housing with proper strain relief for outdoor use.

📡 Multi-pet support

Track multiple collar tags at once — show per-pet data on the same display.

📊 Richer display feedback

Add more visual output (e.g. e-ink or extra LEDs) without increasing power much.